Phong Illumination
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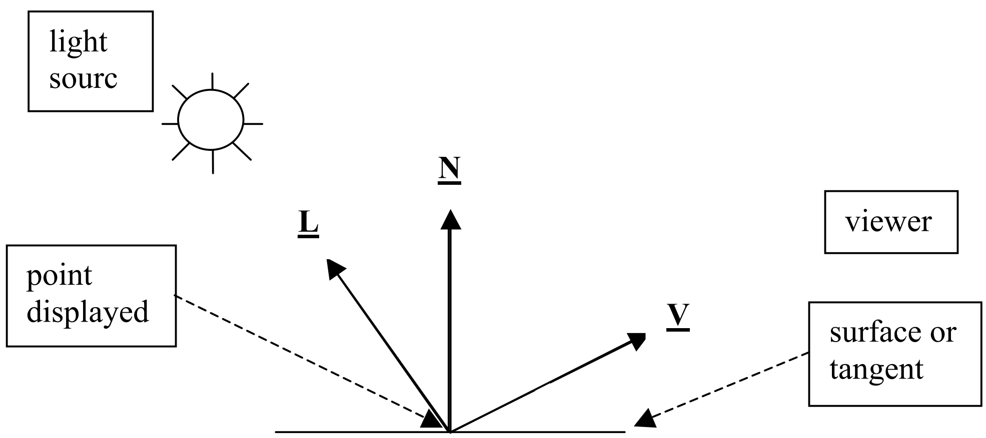
1. Overview of Illumination models

There are two main components to displaying an illuminated object, an illumination model which determines the color of individual points on an object, and a shading model which determines how to average these colors across a surface so that it looks smooth, flat, bumpy, or whatever. In these notes we'll just be looking at the illumination model. We'll assume that we have a viewer who is looking at the point on the surface from a known location, that we have a light source with known intensity, color, and location, that we can compute the normal to the surface of the object at the point, and that we know the color and shininess of the surface of the object. So the basic setup is as shown in the following figure:



L and **V** are unit vectors pointing towards the light source and viewer, respectively, and **N** is the unit normal vector. We can easily compute **L** and **V** by subtracting known points, and we use standard techniques to find the normal vector **N**. If the point is on a plane this is trivial, and if it is a point on a curved surface then the normal is to the tangent plane at that point. We'll assume throughout that we have already performed the calculations to determine that the point is visible to both the light source and the viewer.

The intensity of the light source will be named I_l. Initially we'll ignore the drop in intensity that occurs when you get farther from the light (this is called attenuation), and will then account for it later in these notes.

For most of these notes we'll assume that we are dealing with a single light wavelength so that we can develop the formulas without having to keep dealing with (R, G, B) triples. Then we will generalize the results to the RGB model.

In these notes we'll be describing the Phong illumination model. There are other models, but Phong dominates, and it is the model supported by most graphics software systems. The primary alternative is the Torrance-Sparrow model, first described by Blinn, which is based of the physics of reflection. By comparison, the Phong model isn't based on much physical reality, but it works pretty well and is much less complicated than Torrance-Sparrow.

2. Ambient, Diffuse, and Specular Components

The Phong illumination model has three components, ambient, diffuse, and specular. They will be computed separately, and give rise to three intensity (color) values, I_{amb}, I_{diff}, and I_{spec}. The total intensity at the point will be the sum of these three components,

$$I = I_{amb} + I_{diff} + I_{spec}$$

Ambient light is background light that can't be attributed to any specific light source. Without it scenes look artificial because, for example, a side of an object that doesn't face a single light source will be completely invisible since it won't have diffuse or specular intensities.

Diffuse light is scattered light from a light source. It is the same in all directions, so it doesn't depend on the viewer's location, but only on the position of the light source relative to the surface. If the light source is directly above the surface there will be more diffuse light than if the light is coming in at an angle to the surface.

Specular light is the hot spot. If you take a shiny object like a waxed apple out into bright sunlight, then it will be colored by diffuse and ambient light. But there will also be a shiny hot spot which is around the reflection of the sun in the apple from your viewpoint.

3. Ambient Light

The ambient component of the illumination model is the simplest. Since it just represents light that is everywhere in the scene, and isn't attributable to any light source, it is just a constant that is added into every computed intensity. A typical intensity is something like $I_a = 0.2$ or 0.3. Try a number like this and then adjust up or down until the image meets your needs.

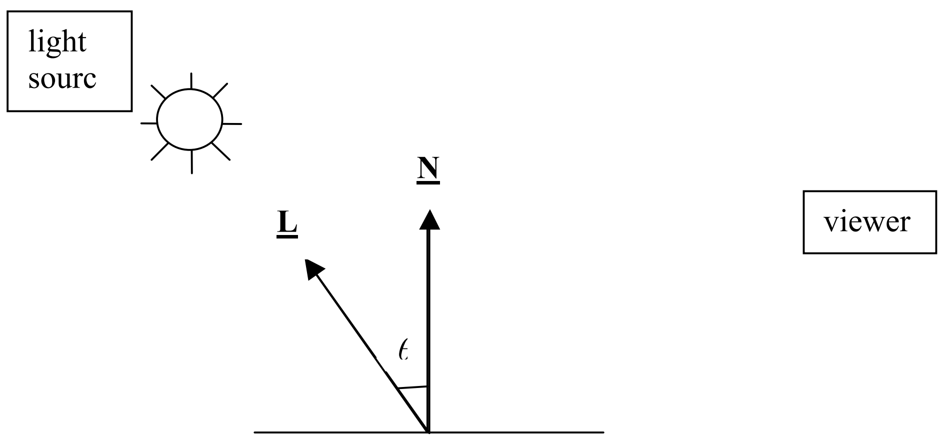
Different surfaces will reflect back different fractions of this light, depending on surface properties and color. We'll use a constant k_{amb} called the ambient reflectance coefficient, which satisfies $0 \leq k_{amb} \leq 1$, and represents the fraction reflected back for the current surface and wavelength. This constant basically represents the color of the surface, and so, for example, a yellow surface will have k_{amb} values that are high in the red and green frequencies and low in the blue frequency.

Then the ambient intensity is calculated as:

$$I_{amb} = k_{amb} \times I_{a.}$$

4. Diffuse Light

This is scattered light, which only depends on the position of the light source relative to the plane that we are displaying. If the light source is straight above the plane (e.g., the sun in the middle of the day) then we get the most scattered light. If it is coming in at an angle that just grazes the plane (e.g., sunset) then we get the least amount of scattered light. So looking at the figure below, where θ is the angle between **L** and **N**, what we need is a function that ranges from 0 to 1 and is at its maximum when $\theta = 0°$, and approaches zero as θ approaches 90°. Cosine does this for us, and so the diffuse model includes a multiplier of $\cos(\theta)$, which is **L**.**N**.



Now it is just a matter of filling in some details. The scattered light also depends on the intensity and color of the light source, I_l, and on the color of the surface, which will determine how much of the light at this frequency (color) will be reflected back off the surface and not be absorbed. This will be modeled by k_{diff}, the diffuse reflectance coefficient where $0 \leq k_{diff} \leq 1$. Putting all this together, the diffuse intensity will be defined by

$$I_{diff} = I_l \times k_{diff} \times (\underline{\mathbf{L}}.\underline{\mathbf{N}}).$$

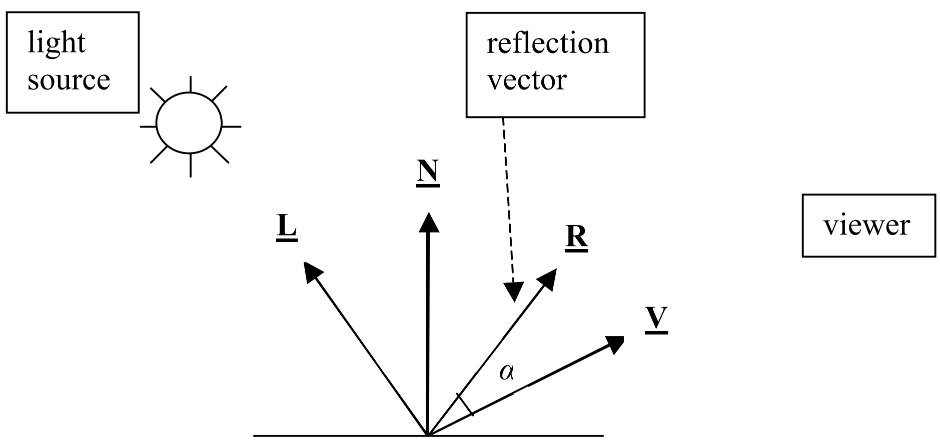
Since k_{amb} and k_{diff} are dependant on surface color, they are usually set to the same value.

5. Specular Light

Although the whole illumination model is usually named after Phong, he is primarily responsible for the specular component. The basic idea is that in addition to light source intensity and color the hot spot is based on two things; the shininess of the object and how close the viewer is to looking down the pure reflection vector. A shiny object (e.g., obsidian or a waxed apple) will have a small hot spot, whereas a less shiny object (e.g., my keyboard) will have larger hot spots.

First we need a bit more detail in out illumination diagram. We've added a new unit vector, **R**, which is a pure reflection vector. I.e., the angle between **L** and **N**, which we called θ in the previous figure, is also the angle between **R** and **N**. α is our name for the angle between **R** and **V**.

For the rest of this discussion we'll assume that $|\alpha| < 90°$ (so **R**.**V** > 0). If it isn't then the specular component is just set to 0.



As we discussed above, the hot spot should only exist if α is very small. It would be nice to just use the dot product again, and use cos(α) to model this, but unfortunately cosine drops off in value relatively slowly. E.g., if **R** and **V** were 60° apart, we would still get half (cos(60°)) of the intensity that we would get if the viewer were looking straight down the reflection vector. To avoid this slow drop off, Phong uses $\cos^n(\alpha)$ for large enough n, which gets the effect that we want. If α is small, and so its cosine is close to 1, then taking it to the n'th power (for large n) will still give a value close to 1. If α were, say, 60°, and n were 20, then its cosine's n'th power would be about

0.000001, which means that there would be no specular effect. So the multiplier in Phong's specular model is $(\underline{\mathbf{R}}.\underline{\mathbf{V}})^n$, which gives the cosine of α to the n'th power. This leads to the specular intensity equation

$$I_{spec} = I_l \times k_{spec} \times (\underline{\mathbf{R}}.\underline{\mathbf{V}})^n,$$

where $0 \leq k_{spec} \leq 1$ is the specular reflectance coefficient. With diffuse light the reflectance coefficient depended on the color of the surface. However if you look at a shiny object in a bright light (e.g., a waxed apple in the sunlight) you'll see that the color of the hotspot is closer to the color of the light source, not of the object. It also depends on θ, but that is usually ignored giving a constant k_{spec}. So select a k_{spec} that mainly depends on the color of the light source, but also includes some component from the surface color.

The main thing that you need to remember when using a Phong illumination model system is that increasing the size of n will decrease the size of the hot spot, and *vice versa*. So shiny surfaces should have bigger n values, dull surfaces should have smaller n values. Try a number like 50, and then adjust it up or down until you are happy with the image.

50 might seem like a large power to use, but the shape of the cosine function is so flat near 0° that you need to use a large power. To show why, I've put together a table of powers of cosines for angles in the range that are relevant.

	$\alpha = 5°$	$\alpha = 10°$	$\alpha = 15°$	$\alpha = 20°$
$\text{Cos}(\alpha)$	.9962	.9848	.9659	.9397
$\text{Cos}^{10}(\alpha)$	.9626	.8581	.7070	.5369
$\text{Cos}^{20}(\alpha)$	.9266	.7363	.4999	.2882
$\text{Cos}^{30}(\alpha)$	.8919	.6317	.3534	.1547
$\text{Cos}^{40}(\alpha)$	.8586	.5421	.2499	.0831
$\text{Cos}^{50}(\alpha)$	.8264	.4651	.1767	.0446
$\text{Cos}^{60}(\alpha)$	.7955	.3991	.1249	.0239
$\text{Cos}^{70}(\alpha)$	.7658	.3425	.0883	.0129

So another approach to selecting a value for n, as compared to just trying 50 and adjusting, is to use this table. E.g., if you want a hot spot that is less that 10% of central intensity by the time that you are 15° from the center, then you'll need to an exponent close to 70.

6. Attenuation

As you get further from a light source, the intensity drops off by the square of the distance. This is called attenuation. If the light source is at about the same distance to all of the objects in the scene, then attenuation can often be ignored, but in many cases you must account for it. E.g., say that you are building a night view of a city, lit by street lights. You won't want the illumination from a street light in one part of the city to be just as bright as it is a few miles away in a different part of the city.

Most graphics system let you specify an attenuation function that is based on the distance, *d*, of the light source to the object with the formula

$$f_{att}(d) = (a + bd + cd^2)^{-1}$$

and then use $f_{att}(d) * I_l$ as the intensity of the light source at a distance of d from its location.

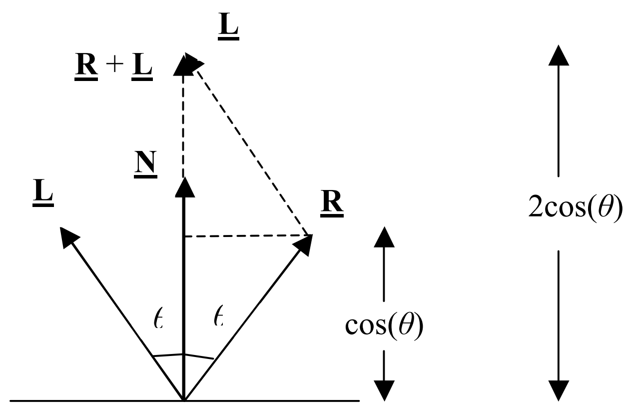
In practice it is usually sufficient to ignore the squared term (although this really offends physicists) by setting $c = 0$. Also, when $d = 0$ it makes sense for $f_{att}(d)$ to be 1, so this implies that a should always be set equal to 1. Another factor to consider is a light source at infinity or great distance. In that case we want to ignore attenuation and so we just set $f_{att}(d)$ to 1.

Incorporating this into the overall model we get the illumination formula:

$$I = I_{amb} + f_{att}(d)\, \mathrm{I_l}\, (k_{diff}\, (\underline{\mathbf{L}}.\underline{\mathbf{N}}) + k_{spec}\, (\underline{\mathbf{R}}.\underline{\mathbf{V}})^n).$$

7. Computing the reflection vector

To compute the unit vector **R** from our known quantities, **L** and **N**, we need the figure below.



Here we have added **L** to **R**, which gives the vector **R** + **L** extending up in the direction of **N**. Looking at the bottom right triangle, its height is cos(θ) since the length of **R** is 1 (unit vector), and so the vector **R** + **L** has the same direction as the unit vector **N**, with length 2cos(θ). I.e.,

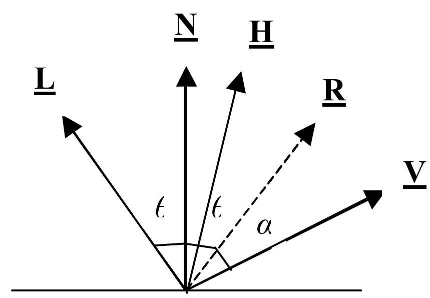
$$\mathbf{R} + \mathbf{L} = 2\cos(\theta)\,\mathbf{N}.$$

Using the dot product **L**.**N** for cos(θ), this gives us an equation for **R**,

$$\mathbf{R} = 2(\mathbf{L}.\mathbf{N})\mathbf{N} - \mathbf{L}.$$

8. Using the Half Angle Vector Instead of the Reflection Vector

If we have to run the Phong illumination model on a lot of points (which is typical) then computing the reflection vectors as described in the last section is relatively expensive. So the common approach is to use the half angle vector, **H**, which is the vector half way between **L** and **V** as shown in the figure below which, as we'll see, is a hack, but gives a good enough approximation of specular effects.



Phong's basic idea was that $\cos^n(\alpha)$ is a convenient value which is close to 1 for small α, and then rapidly drops off to 0 as α gets larger. So, we should be able to use $\cos^m(\alpha/2)$ where m is larger than n, since it is even closer to 1 for small α, and also rapidly drops off to 0 as α gets larger, to get just as good an effect. This is what the half angle approach does.

Looking at the figure, the angle between **L** and **V** is $2\theta + \alpha$, and so the angle between **L** and **H** (which is half way between them) is $\theta + \alpha/2$. Since the angle from **L** to **N** is θ, this leaves $\alpha/2$ as the angle between **N** and **H**. So we can replace $(\mathbf{R}.\mathbf{V})^n$ in the illumination model with $(\mathbf{N}.\mathbf{H})^m$, without loss of quality. By convention we use n instead of m, and so the specular component equation becomes:

$$I_{spec} = I_l \times k_{spec} \times (\mathbf{N}.\mathbf{H})^n$$

Of course this only makes sense if computing the unit vector **H** is faster than computing **R**, but we can do this with:

$$\begin{aligned} \mathbf{H} &= (\mathbf{L} + \mathbf{V}) / 2, \text{ normalized} \\ &= \mathbf{L} + \mathbf{V}, \text{ normalized} \\ &= (\mathbf{L} + \mathbf{V}) / |\mathbf{L} + \mathbf{V}| \end{aligned}$$

which is fast, and so this is the approach usually taken.

9. Multiple Light Sources

If a scene has multiple light sources, we just add their diffuse and specular effects. The ambient component will only be included once. So the illumination equation becomes:

$$I = k_{amb}\, I_a + \sum_{l=1}^{n} \{ f_{att}(d_l) I_l\, [k_{diff}(\underline{\mathbf{N}}.\underline{\mathbf{L}}) + k_{spec,l}(\underline{\mathbf{N}}.\underline{\mathbf{H}})^n] \}$$

Note that the specular reflectance coefficient varies by light source since it depends, at least partially, on the color of the source, whereas the diffuse reflectance coefficient and *n* are constant since they only depend on the surface color and properties. I'm assuming that the same attenuation function applies to all light sources but the result will, of course, depend on d_l, the distance from light source *l*.

10. Multiple Frequencies (e.g. RGB)

Until now we've simplified the equations by assuming that we have a single frequency, which will only apply in the rare case when we are doing gray scale images, while usually we want to be dealing with the RGB system.

To change to RGB the light intensities and the *k* values will all become triples.

The light source and the ambient light will be represented as $I_l = (I_{lR}, I_{lG}, I_{lB})$ and $I_a = (I_{aR}, I_{aG}, I_{aB})$, respectively. In most cases light sources and ambient light are white light, in so the three components will usually be equal,

The *k* values will also be triples:

$k_{amb} = (k_{ambR}, k_{ambG}, k_{ambB})$,
$k_{diff} = (k_{diffR}, k_{diffG}, k_{diffB})$, and
$k_{spec} = (k_{specR}, k_{specG}, k_{specB})$.

As we discussed earlier, usually the k_{amb} and k_{diff} triples will be identical.

Tying this together you can either consider the final illumination equation to be over a set of triples, or (and this is probably easier) as three equations with one equation for each primary color.